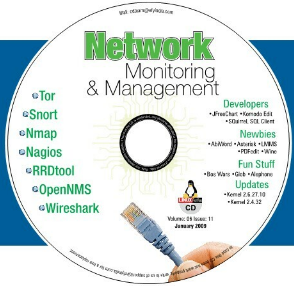

Essential Networking Tools

This month's CD packs in a variety of network tools that you can try out, and which may help you with your network admin workload

Network management, monitoring and security are some of the additional workload you get stuck with when you're in charge of a network. This month's *LFY* CD has a few tools that may come in handy.

Nmap, or the Network Mapper, is a utility for network exploration or security auditing. It is useful for tasks such as network inventory, managing service upgrade schedules, and monitoring host or service uptime. Nmap uses raw IP packets to determine what hosts are available on the network, the services (application name and version) those hosts are offering, the operating systems (and OS versions) they are running, the type of packet filters/firewalls in use, and dozens of other characteristics. It was designed to rapidly scan large networks, but works fine against single hosts. Nmap runs on all major computer operating systems, and both console and graphical versions are available.

/software/powerusers/nmap

OpenNMS is the world's first enterprise-grade network management platform. OpenNMS focuses on three main areas: *Service Polling* that determines service availability and reporting on it; *Data Collection* that collects, stores and reports network information as well as generates thresholds; and *Event and Notification Management* that is responsible for receiving events, both internal and external, and using those events to feed a robust notification system, including escalation.

/software/powerusers/opennms

RRDtool, or the Round Robin Database tool, is an industry-standard, high-performance data logging and graphing system for time series data. It can be used to write your custom monitoring shell scripts or create whole applications using its Perl, Python, Ruby, TCL or PHP bindings.

/software/powerusers/rrdtool

Nagios is an enterprise-class network monitoring tool. It allows you to gain insight into your network and fix problems before users know they even exist. It's stable, scalable, supported, and extensible. Most importantly, it works!

/software/powerusers/nagios

Snort is a network intrusion prevention system, capable of performing real-time traffic analysis and packet logging on IP networks. It can perform protocol analysis, content searching/matching and can be used to detect a variety of attacks and probes, such as buffer overflows, stealth port scans, CGI attacks, SMB probes, OS fingerprinting attempts, and much more.

Snort uses a flexible rules language to describe traffic that it should collect or pass, as well as a detection engine that utilises a modular plug-in architecture. Snort has a real-time alerting capability as well, incorporating alerting mechanisms for syslog, a user specified file, a UNIX socket, or WinPopup messages to Windows clients using Samba's smbclient.

Snort has three primary uses: it can be used as a straight packet sniffer like *tcpdump*, a packet logger (useful for network traffic debugging, etc), or as a full-blown network intrusion prevention system.

/software/powerusers/snort

Tor is a network of virtual tunnels that allows people and groups to improve their privacy and security on the Internet. The aim of Tor is to improve your privacy by sending your traffic through a series of proxies. Your communication is encrypted in multiple layers and routed via multiple hops through the Tor network to the final receiver. Tor works with many of your existing applications, including